

VIRTUAL STUDY MONITORING AND EVALUATION

Monitoring Plan, KPIs, Quality Control, and Simulated Dashboard

Virtual RCT of a Smartphone-Delivered CBT Application for Mild-to-Moderate Generalized Anxiety in Adults

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Week 5 Task: Virtual Study Monitoring and Evaluation

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1. Introduction and Purpose

This document is the fifth companion to the Week 1–4 deliverables for study VCR-2026-001 and defines the monitoring and evaluation framework the study team would use to maintain oversight of study progress, data quality, and protocol adherence throughout conduct. Because this is a fully decentralized trial, monitoring cannot rely on a Clinical Research Associate visiting a physical site to review paper source documents; instead, oversight is performed centrally, continuously, and remotely, using the same electronic systems described in the Week 2 Data Collection and Management Strategy.

This plan defines what gets monitored, how often, by whom, against what numeric thresholds, and shows a simulated monitoring dashboard built from the same underlying study data used in the Week 4 analysis report, so the KPIs below are illustrated with realistic, internally consistent figures rather than placeholder numbers.

2. Key Monitoring Areas

Four areas require regular monitoring throughout the study, consistent with ICH E6(R2) §5.18 and the risk-based monitoring approach introduced in the Week 3 compliance framework.

Monitoring Area	What Is Tracked	Primary Risk if Unmonitored
Participant Adherence & Retention	Enrollment pace, visit-window compliance, dropout/attrition rate	Underpowered analysis; biased results if dropout is differential by severity
Data Quality	Missing data, query volume and resolution time, EDC/app data integration accuracy	Invalid or unusable data at analysis; delayed database lock
Regulatory Compliance	IRB approval currency, consent completeness, Part 11 audit-trail integrity	Non-compliance findings during audit; potential IRB suspension
Participant Safety	Adverse event reporting timeliness, suicide-risk flag response time, protocol deviations	Delayed safety response; harm to participants; regulatory reporting failure

3. Monitoring Schedule, Roles, and Responsibilities

Monitoring activities are scheduled at three cadences — continuous (automated), weekly, and periodic — each with a named owner.

Activity	Frequency	Responsible Role
Automated safety flag monitoring (PHQ-9 item 9)	Continuous (real-time)	Automated system → Study Clinician
Data query review and resolution	Weekly (target: <5-day resolution SLA)	Data Manager

Activity	Frequency	Responsible Role
Enrollment and retention tracking	Weekly	Study Coordinator
Centralized risk-based record review (10% sample)	Biweekly	Data Manager (per Week 3 monitoring log)
Adverse event log review	Biweekly	Study Clinician / Principal Investigator
Protocol deviation log review	Monthly	Regulatory/QA Lead
Dashboard refresh and circulation to study team	Weekly	Data Manager
Formal monitoring report to IRB	Quarterly	Principal Investigator
Full compliance self-audit (per Week 3 checklist)	Quarterly	Regulatory/QA Lead

4. Key Performance Indicators and Quality Control

The following KPIs are tracked on the monitoring dashboard (Section 5) and reviewed against explicit numeric thresholds so that any drift is caught early rather than discovered at database lock.

KPI	Target Threshold	Escalation Trigger
Enrollment vs. target pace	$\geq 95\%$ of planned cumulative enrollment by each recruitment week	2 consecutive weeks below 90% of target pace
Retention through Week 12	$\geq 80\%$ (per Week 1 power calculation assumption)	Projected retention falls below 75% at any interim look
Data query resolution within SLA	$\geq 85\%$ of queries resolved within 5 business days	SLA compliance below 80% for 2 consecutive weeks
Median query resolution time	≤ 3 business days	Median exceeds 5 days
Protocol deviations (major)	0 tolerated without CAPA	Any major deviation triggers immediate CAPA (Section 7)
Serious AE reporting timeliness	100% reported to IRB within 24 hours	Any SAE reported >24 hours triggers root-cause review
EDC–analytics data reconciliation mismatch	0 unresolved mismatches after nightly sync	Any mismatch open >48 hours

Quality control is implemented at three levels: (1) automated, point-of-entry validation in the EDC (Week 2 §5), (2) the KPI thresholds above, reviewed on the cadence in Section 3, and (3) the quarterly compliance self-audit against the 15-item checklist established in the Week 3 deliverable.

5. Simulated Monitoring Dashboard

The dashboard below was generated from simulated study data (the same synthetic dataset used in the Week 4 analysis report, extended with simulated operational metrics for enrollment, query tracking, adverse events, and protocol deviations) to demonstrate how the KPIs in Section 4 would be visualized in practice.

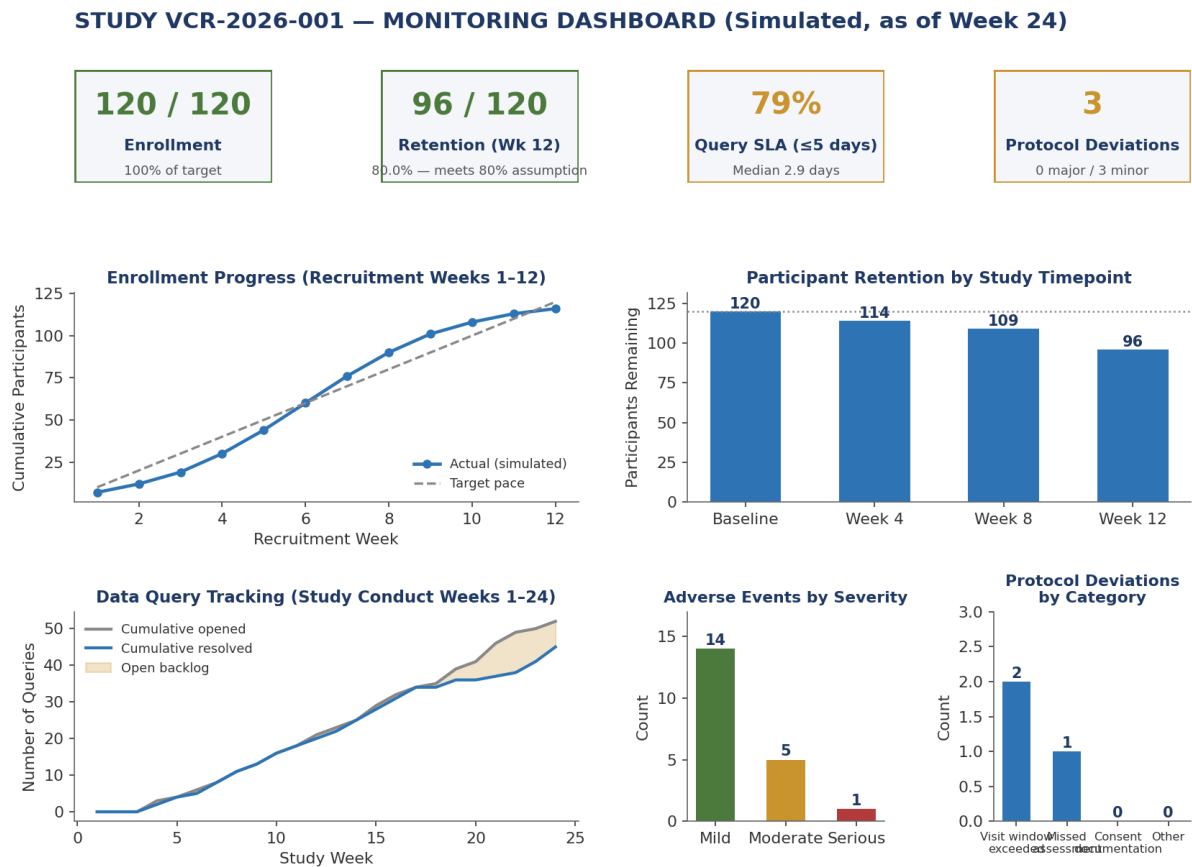


Figure 1. Simulated monitoring dashboard, snapshot as of Study Week 24. KPI cards (top) are colored green when the target threshold is met and amber when a metric requires attention.

5.1 Reading the Dashboard

- Enrollment reached 120/120 (100% of target) by recruitment Week 12, tracking close to the planned pace throughout, with a brief lag around Weeks 4–6 that self-corrected without intervention.
- Retention fell from 120 at Baseline to 96 at Week 12 (80.0%), landing exactly at the threshold assumed in the Week 1 power calculation — acceptable, but with no margin for further loss, so retention should stay a standing agenda item rather than be considered “resolved.”
- Query SLA compliance is 79%, below the 85% target threshold defined in Section 4 — this is flagged amber on the dashboard and is addressed as an open finding in Section 7 (CAPA log) rather than left unaddressed.
- Adverse events show the expected distribution (mostly mild, a small number moderate, one serious event), consistent with the AE severity grading approach established in the Week 3 sample documentation.

- Protocol deviations remain low (3 minor, 0 major) and concentrated in visit-window timing rather than consent or safety-related categories, which is the lower-risk deviation profile ICH E6(R2) risk-based monitoring is designed to tolerate without escalation.

6. Protocol Deviation and Adverse Event Handling

6.1 Protocol Deviation Process

1. Identification: a deviation is identified either automatically (e.g., a visit occurring outside its ± 3 -day protocol window, per the Week 2 validation rules) or manually (reported by the Study Coordinator or participant).
2. Classification: the Regulatory/QA Lead classifies the deviation as minor (no impact on participant safety or data integrity, e.g., a visit window exceeded by 1–2 days) or major (potential impact on safety, rights, or data integrity, e.g., enrollment of an ineligible participant).
3. Documentation: every deviation is logged with a unique ID, date, description, classification, and root cause in the deviation log.
4. Reporting: major deviations are reported to the IRB within 5 business days; minor deviations are summarized in the quarterly monitoring report.
5. Corrective action: every major deviation, and any pattern of ≥ 3 similar minor deviations, triggers a CAPA entry (Section 7).

6.2 Adverse Event Handling

The adverse event process follows the pathway established in the Week 3 sample documentation: onset and report timestamps are captured to verify the 24-hour serious-AE reporting clock, severity and causality are graded by the Study Clinician, and the automated PHQ-9 item-9 suicide-risk flag routes to same-day clinician review independent of the standard AE workflow, ensuring safety-critical signals are never queued behind routine monitoring activities.

7. Corrective and Preventive Action (CAPA) Log

The dashboard's amber-flagged query-SLA metric (Section 5) is used here as a worked example of how a monitoring finding becomes a documented CAPA entry.

CAPA ENTRY — SAMPLE

CAPA ID: CAPA-2026-002

Date opened: 2026-Week 24 monitoring review

Finding: Data query resolution SLA compliance at 79%, below the 85% target threshold; median resolution time 2.9 days (within target) but a growing open backlog visible in Figure 1

Root cause (assessed): Query volume increased around Weeks 18–22 without a corresponding increase in Data Manager review capacity

Corrective action: Redistribute a portion of routine query review to the Study Coordinator under Data Manager supervision during high-volume weeks

Preventive action: Add a rolling 4-week query-volume forecast to the weekly dashboard refresh so capacity can be adjusted proactively

Owner: Data Manager

Target closure date: Within 4 weeks of CAPA opening

Verification method: SLA compliance $\geq 85\%$ sustained for 2 consecutive monitoring cycles before closure

Rationale: rather than treating an amber KPI as a one-line note, routing it through a structured CAPA with a named root cause, action, owner, and closure criterion is what allows an auditor (or the IRB) to see that a monitoring finding was actually acted on, not just observed.

8. References

International Council for Harmonisation. (2016). ICH E6(R2): Good Clinical Practice Guideline, Section 5.18 (Monitoring).

U.S. Food and Drug Administration. (2013). Guidance for Industry: Oversight of Clinical Investigations – A Risk-Based Approach to Monitoring.

Virtual Study Protocol, VCR-2026-001, Week 1 deliverable (sample-size and retention assumptions).

Data Collection and Management Strategy, VCR-2026-001, Week 2 deliverable (validation rules and query SLA definition).

Regulatory Compliance and Documentation, VCR-2026-001, Week 3 deliverable (AE and deviation documentation templates).

Clinical Data Analysis and Reporting, VCR-2026-001, Week 4 deliverable (source dataset for dashboard retention figures).